SALESLENS ENTERPRISE BI PLATFORM • ACADEMIC DELIVERABLE DOC-02

Software Design & UML Architecture Package

Course: B.Tech CSE Semester III • Software Engineering & Project Management • Version 1.0

1. Executive Design Summary & Architecture

SalesLens adopts a **Layered Service-Oriented Model-View-Controller (MVC) Architecture** with an asynchronous **Extract-Transform-Load (ETL) Pipeline-and-Filter Subsystem**.

> **Coupling & Cohesion Architectural Justification**
>
> **Low Coupling:** The Data Cleaning Engine operates independently of the Presentation Layer. Dashboard views interact exclusively with an aggregated Data Access Layer (DAL) through standardized Data Transfer Objects (`DashboardDTO`). If billing schema or region mapping logic changes, dashboard views require zero code modifications.
>
> **High Cohesion:** Each subsystem is single-purposed: `RegionResolutionEngine` resolves geographic anomalies; `QuotaComputationEngine` calculates variances; `DashboardController` manages UI rendering.

2. Use Case Model

The system defines 4 actors (Sales Director, Regional Manager, Data Analyst, IT Admin) interacting with 9 core system use cases:

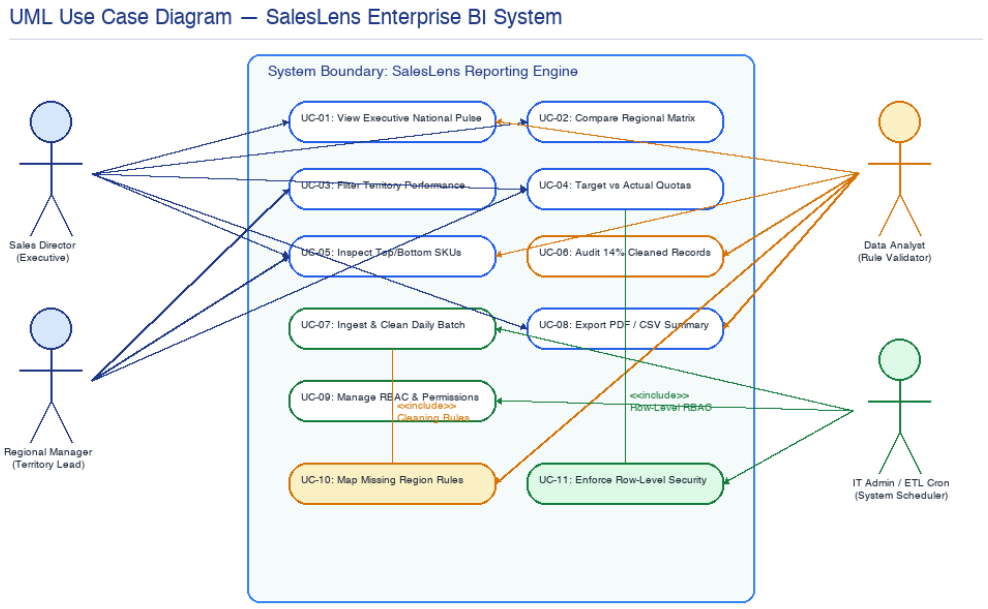


Figure 2.1: UML Use Case Diagram — Actors, System Boundary, and Use Cases

3. Class Diagram (Domain, ETL & Service Layers)

Structured into Domain Entities (`Transaction`, `CustomerMaster`), ETL Subsystem (`RegionResolutionEngine`, `AuditLogger`), and Presentation Controllers (`DashboardController`, `SalesAggregatorService`):

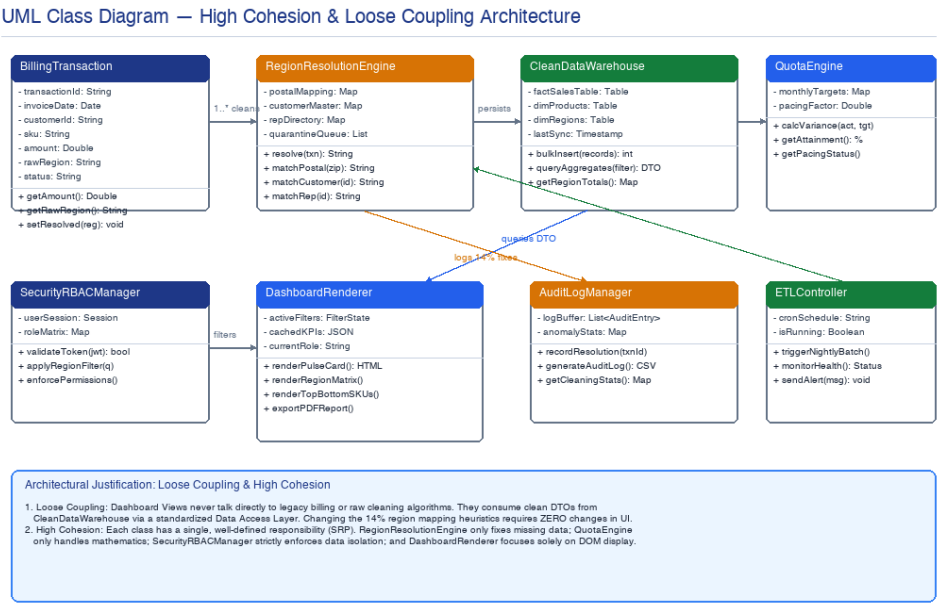


Figure 2.2: UML Class Diagram — Domain Entities, Cleaners, Aggregators & Controllers

4. Behavioral Modeling: Sequence Diagram

Demonstrates the complete sequence from 02:00 AM overnight extraction through 14% cleaning to morning executive UI rendering:

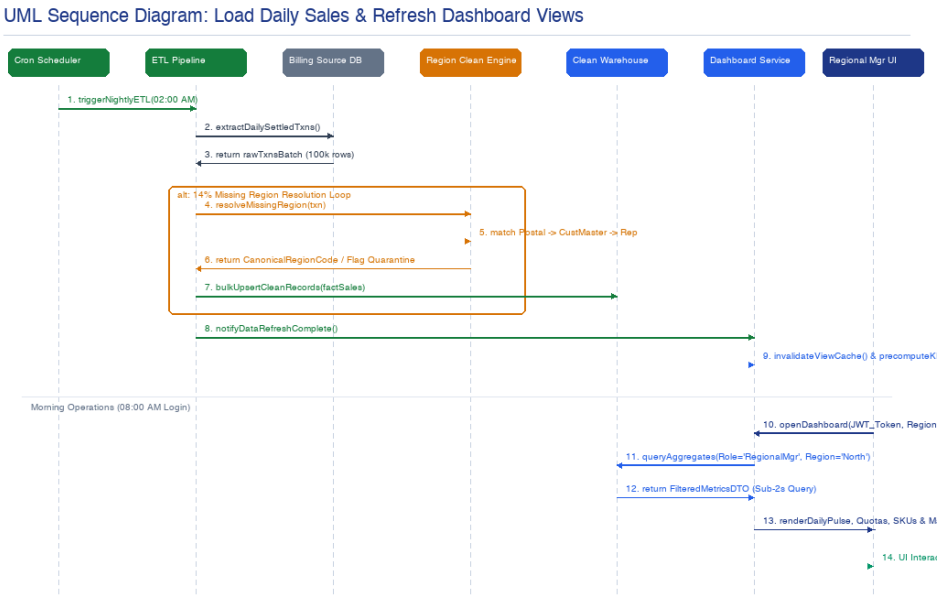


Figure 2.3: Sequence Diagram — Overnight ETL Ingestion & Morning Dashboard Load

5. Behavioral Modeling: Activity Diagram

Illustrates the 4-tier decision waterfall for resolving missing region transactions:

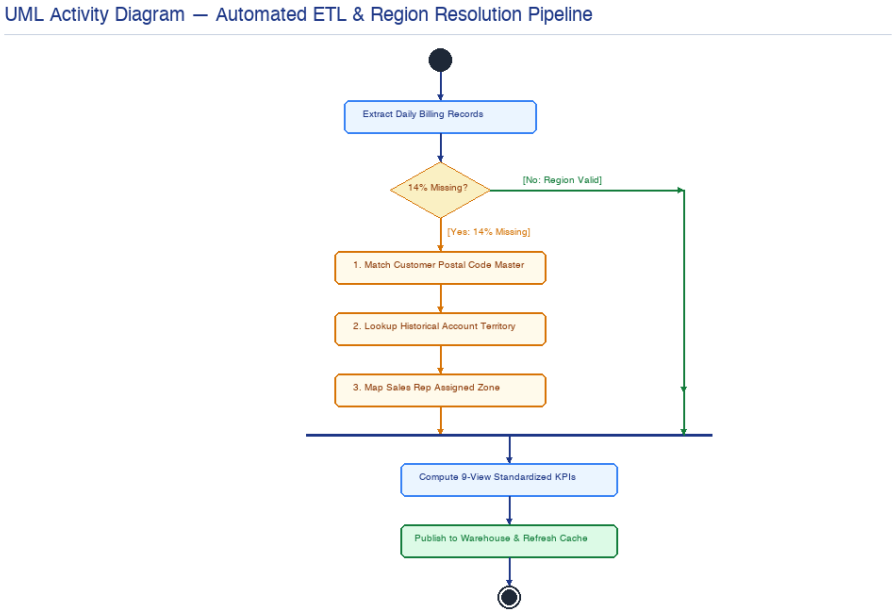


Figure 2.4: Activity Diagram — 4-Tier Waterfall Data Cleaning Pipeline

6. State Machine & Deployment Architecture

Depicts the data load lifecycle with exponential backoff retries and physical hardware tiering:

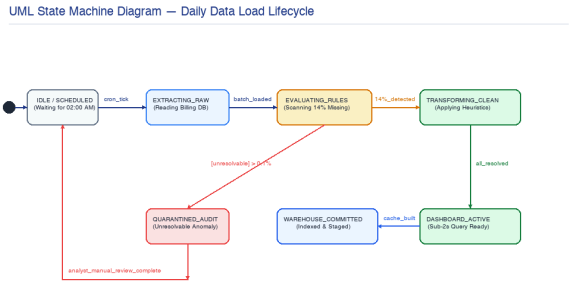


Figure 2.5: State Machine Diagram

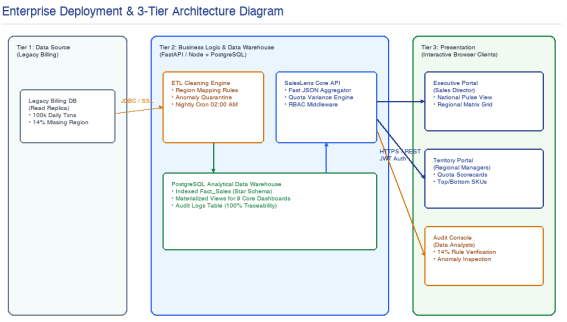


Figure 2.6: Deployment Architecture